

INDIAN STATISTICAL INSTITUTE

Final

M. Math II Year

Algebraic Number Theory

Date: April 30, 2026 Maximum marks 50 Duration: 3 hours

Answer any three questions.

✓ 1. Let $K = \mathbb{Q}(\theta)$, where θ is a root of the irreducible polynomial $T^3 - T^2 - 2T - 8 \in \mathbb{Z}[T]$.

- (a) Let $\theta' = \frac{4}{\theta} \in K$. Show that $\theta' \in \mathcal{O}_K$. [3]
- (b) Prove that $\mathbb{Z} + \mathbb{Z}\theta + \mathbb{Z}\theta'$ is a subring of $\mathcal{O}_K$. [3]
- (c) Show that the discriminant $d(1, \theta, \theta') = -503$. Conclude that $\mathbb{Z} + \mathbb{Z}\theta + \mathbb{Z}\theta' = \mathcal{O}_K$. [6]
- (d) Prove that $\mathcal{O}_K \neq \mathbb{Z}[\alpha]$ for any $\alpha \in \mathcal{O}_K$. [5]

✓ 2. Let K be a number field and let $\mathcal{O}_K$ be its ring of integers. Let $p \in \mathbb{Z}$ be a prime. Prove that $p\mathcal{O}_K \neq \mathcal{O}_K$. Consider the decomposition

$$p\mathcal{O}_K = P_1^{e_1} \cdots P_r^{e_r},$$

where $P_i \in \text{Spec}(\mathcal{O}_K)$, ($P_i \neq P_j$). Let $f_i := [\mathcal{O}_K/P_i : \mathbb{Z}/p\mathbb{Z}]$ for $i = 1, \dots, r$. Prove that [17]

$$e_1 f_1 + \cdots + e_r f_r = n.$$

3. (a) Compute the class number of $\mathbb{Q}(\zeta)$, where ζ is a primitive 7-th root of unity. [5]

(You may use the Minkowski bound $M = \frac{n!}{n^n} \left(\frac{4}{\pi}\right)^s \sqrt{|d_K|}$).

(b) Compute the class number of $\mathbb{Q}(\sqrt{10})$. [7]

(c) Let $K = \mathbb{Q}(\sqrt{-3})$. Find a prime ideal factorization of $7\mathcal{O}_K$ in $\mathcal{O}_K$. [5]

✓ 4. (a) Let $p \in \mathbb{Z}$ be an odd prime and K be a quadratic number field of discriminant d . Show that d is a quadratic residue modulo p if and only if there is a prime ideal P of $\mathcal{O}_K$ such that

$$p\mathcal{O}_K = P \bar{P},$$

and $P \neq \bar{P}$, where $\bar{P}$ is the conjugate of P . [10]

(b) Let K be a number field and d be its discriminant. Prove that d is 0 or 1 modulo 4. [7]